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**B.Sc. (E.C.S.) (Semester - II) (NEP CBCS) Examination:
October/November - 2025
Discrete Mathematics (G06-0204)**

Day & Date: Thursday, 13-11-2025
Time: 03:00 PM To 04:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.
3) Use of non-programmable any type of calculator is allowed.

Q.1 Choose the Correct Alternative.

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- 1) If $f(x) = 3x^2 + 4x - 12$ then $f(-2) = \underline{\hspace{2cm}}$

a) -8 b) 8
c) 0 d) 20
- 2) Order of recurrence $a_r - 4a_{r-3} + 3a_{r-2} = 0$ is _____

a) 1 b) 2
c) 3 d) 0
- 3) A relation R which is reflexive, _____ and transitive is known as equivalence relation.

a) asymmetric b) partially
c) anti symmetric d) symmetric
- 4) If $|A| = 100, |B| = 94, |C| = 65, |A \cap B| = 32, |A \cap C| = 19,$
 $|B \cap C| = 41,$ and $|A \cap B \cap C| = 27$ then $|A \cup B \cup C| = \underline{\hspace{2cm}}$

a) 194 b) 140
c) 258 d) 190
- 5) If set A has 3 elements and set B has 4 elements then number of elements in $A \times B$ are _____ and number of possible relations from A to B are _____ respectively.

a) 7 and 2^7 b) 12 and 2^{12}
c) 7 and 2^{12} d) 2^{12} and 12
- 6) If every element of set A is related to unique element of set B then such relation is called as _____ from the set A to B .

a) void relation b) universal relation
c) function d) None of these

Q.2 Attempt any three of the following.**06**

- Write characteristic equation for the recurrence relation $a_r - 8a_{r-1} + 16a_{r-2} = 0$ Also find characteristic roots.
- State principle of mutual inclusion - exclusion for three sets A, B and C.
- Define one one function and bijective function.
- Let $A = \{a, b, c, d\}$. Let $R = \{(a, a), (a, b), (b, d), (b, a), (c, a), (d, c), (c, b), (a, d), (d, d), (d, c), (c, d)\}$ be the relation defined on the set A. Write $M(R)$, Also draw diagram of R.
- Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8\}$ then find $A \times B$ and $B \times A$. Is Cartesian product commutative?

Q.3 Attempt any two of the following.**06**

- Let $U = \{a, b, c, d, e, f, g, h, i, j, k, l, m, n\}$. Let $A = \{a, b, d, f, g, m, n\}$, $B = \{a, b, c, e, h, i, m\}$ be the subsets of U then write the following sets and their cardinalities $A \cup B$, $A - B$, $B - A$ and B' .
- Let $f: Q \rightarrow Q$ be a function defined by $f(x) = \frac{5x-2}{7}$; for all $x \in Q$. Show that f is bijective.
- State and prove principle of mutual inclusion - exclusion for two sets.

Q.4 Attempt any two of the following.**06**

- Find homogeneous solution of the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 0$
- Define equivalence relation, universal relation, and transitive relation.
- How many integers between 1 to 798 which are divisible by either 3 or 7?

Q.5 Attempt any one of the following.**06**

- Define transitive closure. Hence by using Warshall's algorithm find transitive closure of relation of relation $R = \{(a, c), (c, b), (b, a), (d, b), (d, d)\}$ defined on $A = \{a, b, c, d\}$
- Define particular solution of a recurrence relation. Hence find the particular solution of the recurrence relation $a_r - 4a_{r-1} + 4a_{r-2} = 4^r$